ECE276C Assignment 1: Robot Kinematics and Simple Control

Code: `arm.py` (kinematics), `a1_solution.py` (PyBullet: Q3, Q5, Q6), `figures.py` (Q1, Q4 figures). Outputs are in `results/`. Run with `conda activate q4; python a1_solution.py; python figures.py`.

1. Arm and body-fixed coordinate frames

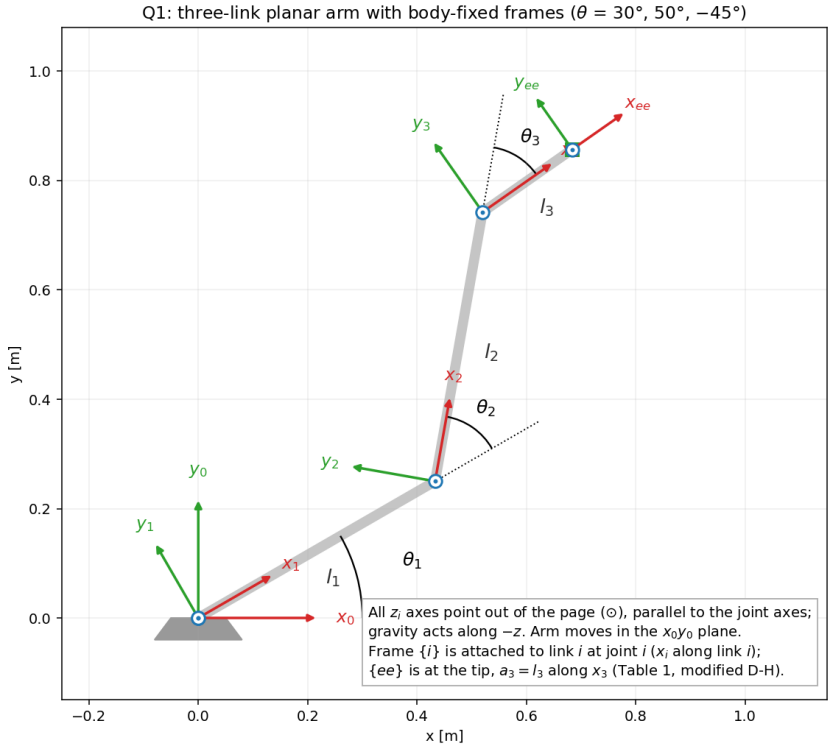


Figure 1: Three-link planar arm with body-fixed frames (modified D-H, Table 1).

Frame $\{0\}$ is fixed to the base. Frame $\{i\}$ $(i = 1, 2, 3)$ is attached to link i with its origin at joint i, x_i along link i and z_i along the joint axis (out of the page, opposite to gravity). Following Table 1, $\{1\}$ is $\{0\}$ rotated by θ_1; $\{2\}$ is translated l_1 along x_1 and rotated by θ_2; $\{3\}$ is translated l_2 along x_2 and rotated by θ_3; $\{ee\}$ is translated l_3 along x_3 with no rotation. At $\theta = (0, 0, 0)$ every x_i is aligned with x_0 and the arm is fully stretched.

2. Forward kinematics

Each D-H row is $T_{i-1,i} = \mathrm{Trans}_x(a_{i-1})\,\mathrm{Rot}_z(\theta_i)$; multiplying the four planar transforms gives

$$\begin{aligned} X &= l_1\cos\theta_1 + l_2\cos(\theta_1+\theta_2) + l_3\cos(\theta_1+\theta_2+\theta_3) \\ Y &= l_1\sin\theta_1 + l_2\sin(\theta_1+\theta_2) + l_3\sin(\theta_1+\theta_2+\theta_3) \\ \theta &= \theta_1+\theta_2+\theta_3 \end{aligned}$$

These equations match the end-effector pose that PyBullet reports for the URDF to within 3×10^{-8} m and 3×10^{-8} rad in all four configurations of Q3.

3. PyBullet renders ($l_1 = l_2 = 0.5$, $l_3 = 0.2$ m)

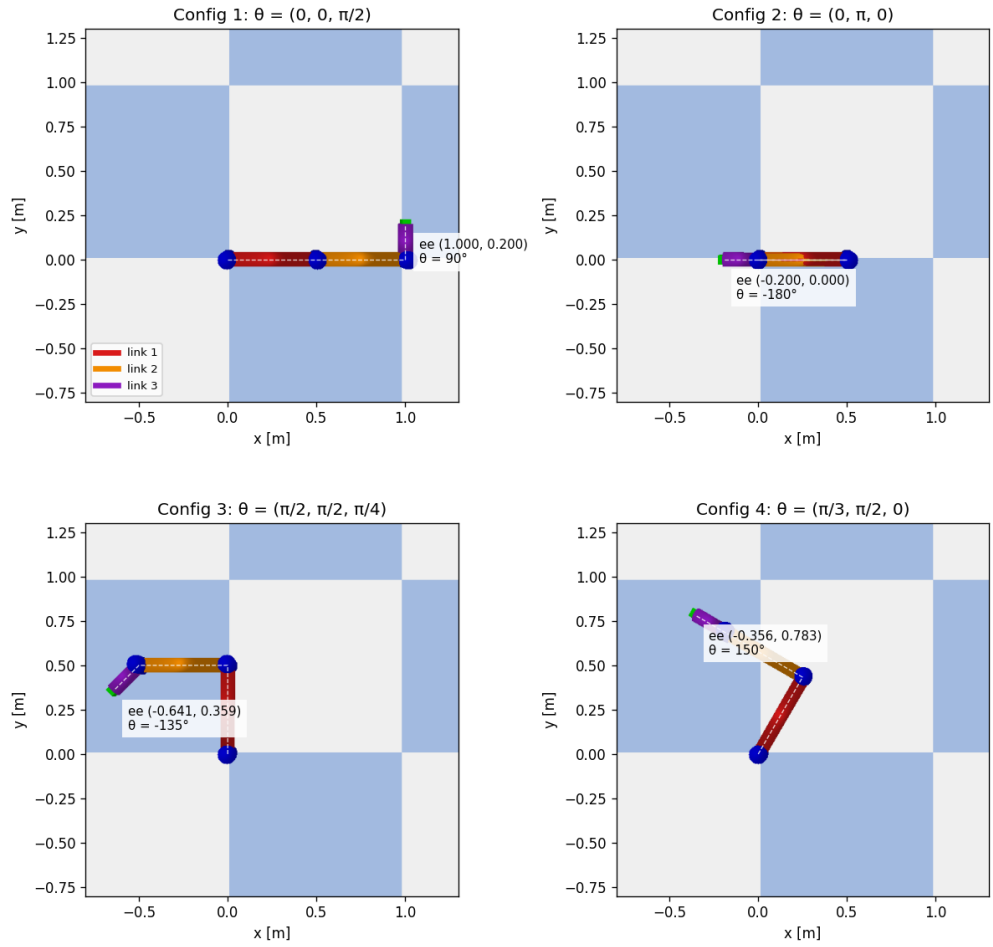


Figure 2: Top-down PyBullet renders (individual images: `results/q3_config{1..4}.png`). Links are recoloured (red, orange, purple) to tell them apart. The white dashed line is the analytic FK skeleton drawn over the render.

Config	$(\theta_1, \theta_2, \theta_3)$	(X, Y) [m]	θ
1	$(0, 0, \pi/2)$	$(1.000, 0.200)$	$90°$
2	$(0, \pi, 0)$	$(-0.200, 0.000)$	$180°$
3	$(\pi/2, \pi/2, \pi/4)$	$(-0.641, 0.359)$	$225°$ $(-135°)$
4	$(\pi/3, \pi/2, 0)$	$(-0.356, 0.783)$	$150°$

In configuration 2, link 2 folds back exactly on top of link 1 ($\theta_2 = \pi$, $l_1 = l_2$), so the two overlap in every view. The overlap shows as mixed red/orange, with link 3 sticking out to $x = -0.2\,\mathrm{m}$.

4. Control block diagrams

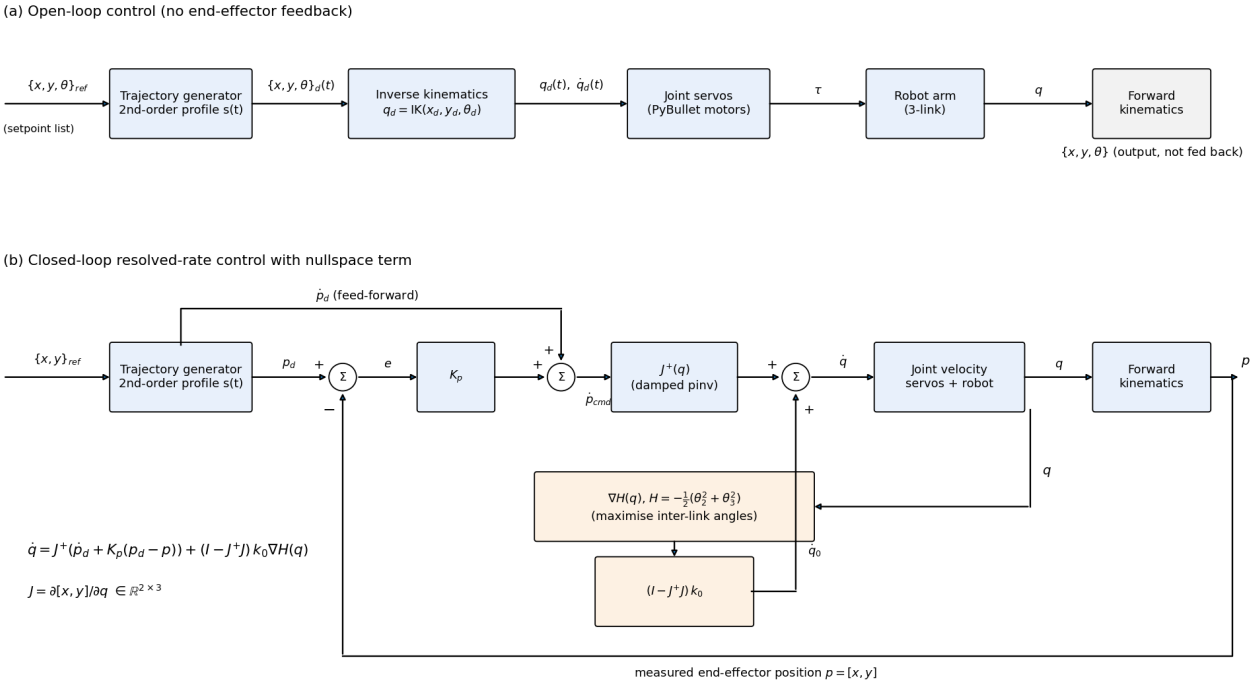


Figure 3: (a) Open loop: the reference is turned into joint setpoints by inverse kinematics, and the end-effector pose is never measured. (b) Closed loop: the measured end-effector position is fed back through a task-space gain and the Jacobian pseudo-inverse, and a nullspace term handles the redundant DOF.

5. Second-order smooth profile and open-loop setpoint control

Profile. A move from $\mathbf{x}_0 = \{x, y, \theta\}_0$ to $\mathbf{x}_{ref}$ over a duration T uses a scalar path parameter:

$$\mathbf{x}_d(t) = \mathbf{x}_0 + s(\tau)\,(\mathbf{x}_{ref} - \mathbf{x}_0), \qquad \tau = t/T, \qquad s(\tau) = \begin{cases} 2\tau^2 & 0 \le \tau < \frac{1}{2} \\ 1 - 2(1-\tau)^2 & \frac{1}{2} \le \tau \le 1 \end{cases}$$

s is piecewise second order (constant acceleration, then constant deceleration): $\dot{s} = 4\tau/T$ and $4(1-\tau)/T$, $\ddot{s} = \pm 4/T^2$. It is C^1, with $s(0) = 0$, $s(1) = 1$ and $\dot{s}(0) = \dot{s}(1) = 0$. The robot therefore starts and stops at each setpoint with zero velocity, and peaks at $2\|\Delta\mathbf{x}\|/T$ halfway. All coordinates share one scalar s, so the end effector moves on the *straight line* between setpoints.

The segment durations are chosen so that every segment has the same peak acceleration ($T_i \propto \sqrt{d_i}$) and the total is 10 s: $T = (2.62, 2.83, 2.02, 2.53)$ s for $1 \to 2 \to 3 \to 4 \to 1$.

Open-loop control. At each 240 Hz step, $\mathbf{x}_d(t)$ is converted to joint setpoints by analytic IK. First the wrist point $(x_w, y_w) = (x - l_3\cos\theta,\ y - l_3\sin\theta)$ is computed. Then

$$\theta_2 = \arccos\frac{x_w^2 + y_w^2 - l_1^2 - l_2^2}{2l_1l_2}, \quad \theta_1 = \operatorname{atan2}(y_w, x_w) - \operatorname{atan2}(l_2\sin\theta_2,\ l_1 + l_2\cos\theta_2), \quad \theta_3 = \theta - \theta_1 - \theta_2.$$

The elbow branch is fixed and angles are unwrapped for continuity. The setpoints $q_d, \dot{q}_d$ go to PyBullet's joint position motors. The end-effector pose is never fed back.

Result (`results/q5_open_loop.gif`, 10 s, 20 fps; the green trail is the end-effector path). In this ideal simulation (exact model, no disturbances) the open-loop scheme tracks well: the maximum position error is 3.8 μm and the maximum orientation error is 7×10^{-4} deg. Every setpoint is reached at zero velocity.

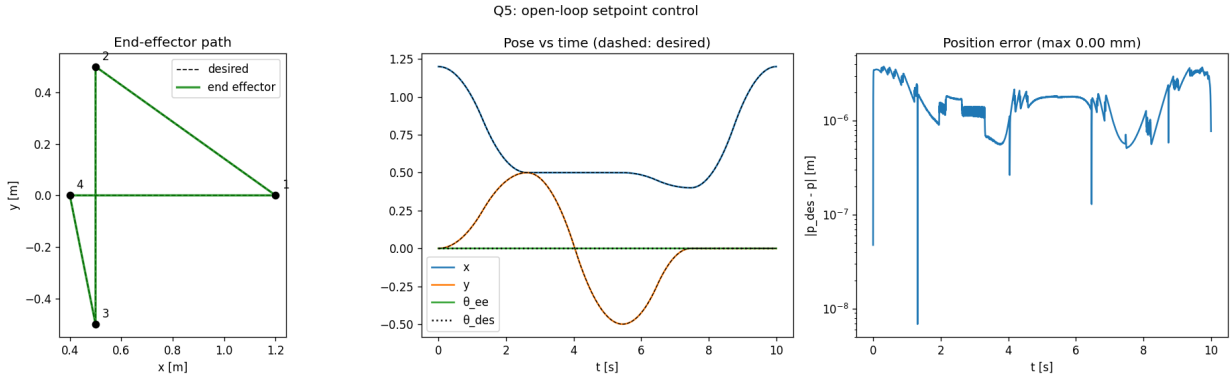


Figure 4: Q5: end-effector path, pose vs. time, and position error.

6. Closed-loop control with a nullspace controller

Only (x, y) is controlled now, so the arm has one redundant DOF. With $p = [x, y]^T$ the *measured* end-effector position and $J(q) = \partial p/\partial q \in \mathbb{R}^{2\times 3}$, the resolved-rate law is

$$\dot{q} = J^+\big(\dot{p}_d + K_p\,(p_d - p)\big) + \big(I - J^+J\big)\,k_0\nabla H(q), \qquad J = \begin{bmatrix} -l_1s_1 - l_2s_{12} - l_3s_{123} & -l_2s_{12} - l_3s_{123} & -l_3s_{123} \\ l_1c_1 + l_2c_{12} + l_3c_{123} & l_2c_{12} + l_3c_{123} & l_3c_{123} \end{bmatrix}.$$

The reference $p_d(t)$, $\dot{p}_d(t)$ is the same straight-line, second-order profile as in Q5. $\dot{q}$ is sent to PyBullet's joint velocity motors, and q and p are read back from the simulator every step. Gains: $K_p = 20\,\mathrm{s}^{-1}$, $k_0 = 10$. The pseudo-inverse is a damped least-squares pseudo-inverse, $J^+ = J^T(JJ^T + \lambda^2 I)^{-1}$ with $\lambda = 10^{-3}$.

Nullspace objective. The angle between link i and link $i+1$ is $\pi - |\theta_{i+1}|$ (it equals π when they are aligned). Maximising both angles is equivalent to driving $\theta_2, \theta_3 \to 0$. A smooth objective for this is

$$H(q) = -\tfrac{1}{2}(\theta_2^2 + \theta_3^2), \qquad \nabla H = [\,0,\ -\theta_2,\ -\theta_3\,]^T.$$

The projector $I - J^+J$ removes every component of this term that would move p, so the x, y trajectory is unaffected. The end-effector orientation θ is left free.

Singularity at location 1. $(1.2, 0)$ is at full reach, so the arm must be fully straight there, which is a singular configuration: no joint velocity moves the tip radially inwards. The simulation therefore starts from a tiny symmetric bend ($\theta_2 = \theta_3 = 0.01$, rotated so the tip stays on the x axis). This picks the elbow direction and gives an initial error of only 0.03 mm.

Result (`results/q6_closed_loop.gif`). The maximum position error is 0.11 mm and the RMS error is 0.06 mm, so the x, y path is followed essentially exactly. Figure 6 compares the inter-link angles with the fixed-$\theta = 0$ solution of Q5, and with a brute-force optimum: for each p_d, a search over all orientations θ for the one minimising $\theta_2^2 + \theta_3^2$. The nullspace controller tracks the optimum closely. It keeps the links much straighter than Q5, whose minimum angle is 23° against 52° here. The bend is spread over both joints, as expected.

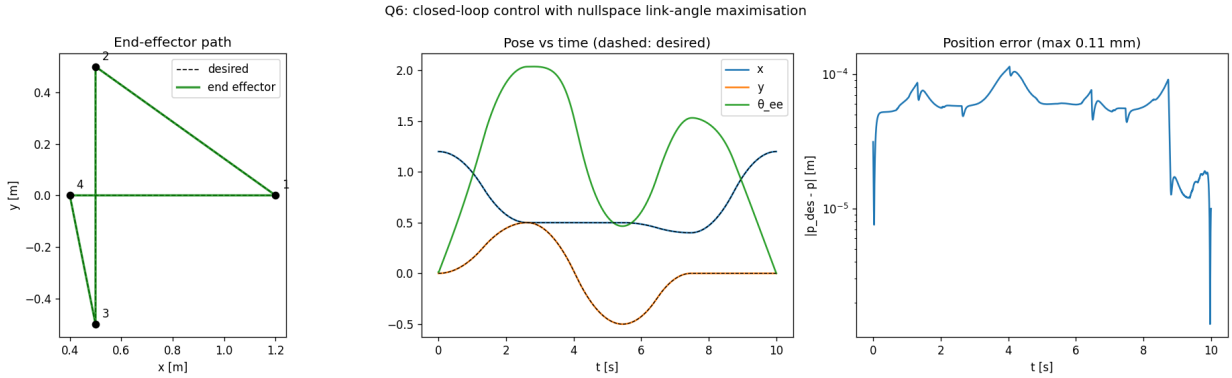


Figure 5: Q6: end-effector path, pose vs. time (the orientation θ_{ee} is free), and position error.

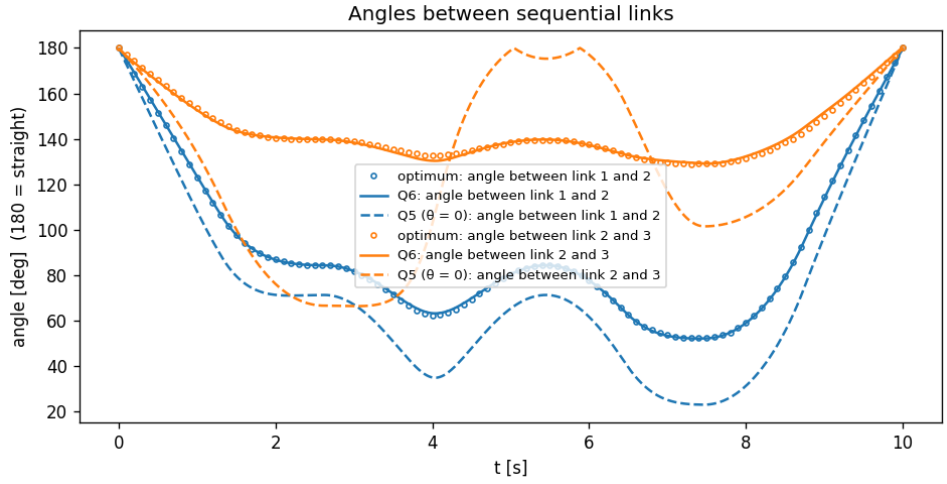


Figure 6: Angles between sequential links: Q6 nullspace controller (solid), the brute-force optimum (circles), and Q5 with $\theta = 0$ (dashed).